Evaluating the Mobility in Pforzheim

In the study course

SUM WP-Modul SUM 3 III.1 Data analysis and visualization

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List of abbreviations

BW – Baden-Württemberg

CSR – Coordinate Reference System

EPSG – European Petroleum Survey Group (code system for coordinate reference systems)

IQR – Interquartile Range

MIV – Motorisierter Individualverkehr (Motorized Individual Transport)

QGIS – Quantum Geographic Information System

UTM – Universal Transverse Mercator (coordinate system)

WMS – Web Map Service

ÖV – Öffentlicher Verkehr (Public Transport)

OSM – OpenStreetMap

WFS – Web Feature Service

1 Introduction

The city of Pforzheim, located in the federal state of Baden-Württemberg, serves as the spatial focus of this seminar paper. Situated approximately 40 kilometers west of Stuttgart, Pforzheim is known for its historical significance in the jewelry industry as well as its role as a regional center of education and employment. With a population of around 135,000 and an area of approximately 98 km^2, the city presents a compact and diverse urban landscape. The aim of this analysis is to evaluate the availability and distribution of mobility options within Pforzheim, using a range of spatial datasets and methods introduced during the course.

This paper applies spatial techniques such as buffer analysis, heatmapping, accessibility zoning, and fragmentation studies to assess the coverage and quality of mobility infrastructure in Pforzheim. While the overall goal is to assess the urban mobility conditions citywide, a special focus is placed on the accessibility of Hochschule Pforzheim and its surrounding districts. This approach allows for a balanced examination of both general mobility conditions and specific spatial inequalities.

2 Description of Used Datasets

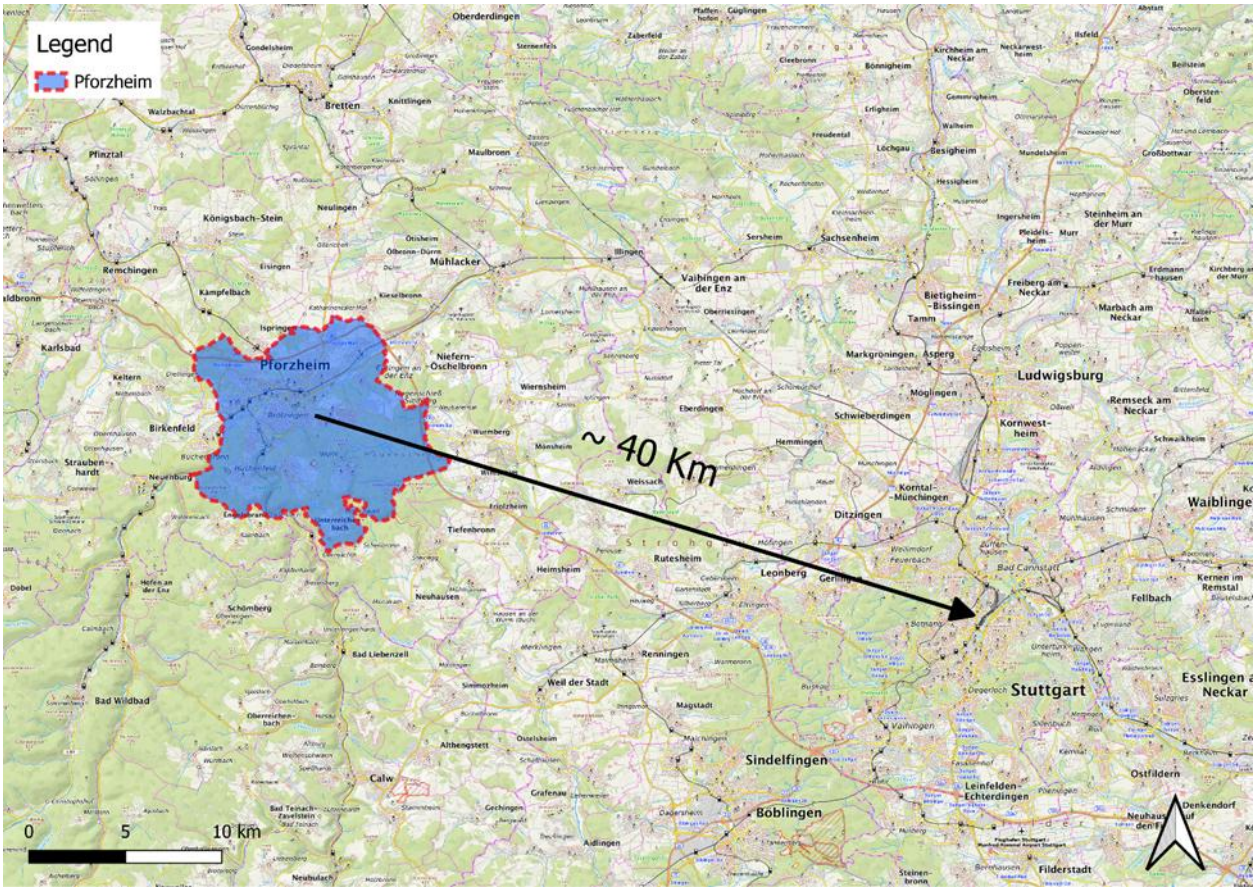
The spatial analysis of mobility in Pforzheim was conducted using openly accessible datasets from multiple sources. All spatial layers were projected in EPSG:25832 (ETRS89 / UTM zone 32N) to ensure consistency in measurements. Below is an overview of the datasets used:

- City boundary and administrative area of Pforzheim: Downloaded from Geofabrik's OpenStreetMap extracts for Baden-Württemberg (Geofabrik GmbH, 2025).
- Street network and bus stops: Extracted from OpenStreetMap via Geofabrik (Geofabrik GmbH, 2025), cleaned and filtered to highlight relevant public transportation infrastructure.
- Bike-sharing and E-scooter sharing station locations: Sourced from MobiData BW's open mobility datasets (MobiData BW, 2025a; 2025b).
- Cycle path layers: The official cycle networks, including RadnetzBW and Kreisnetz, were obtained from MobiData BW (2025c).
- Walking accessibility analysis (10-minute walk from city center): Generated using QGIS network analysis tools based on an estimated pedestrian speed of ~4 km/h.
- Mobility zone analysis: A 200-meter buffer was applied around bus stops, bike-sharing, and e-scooter sharing stations to delineate areas with access to all the three mobility services.
- Heatmap layer: Created with QGIS's Heatmap tool using combined point layers for bike and e-scooter sharing to highlight areas of service concentration.
- University locations: Manually georeferenced using public campus information for Hochschule Pforzheim.
- Basemaps: Most maps were visualized using OpenStreetMap tiles, with some layouts using the TopoPlus Open WMS service from the "Bundesamt für Kartographie und Geodäsie" (Bundesamt für Kartographie und Geodäsie, 2023).

All calculations and spatial processes were performed in QGIS. Any data manipulations not derived from external datasets are indicated as "own measurement."

3 General Overview of Pforzheim

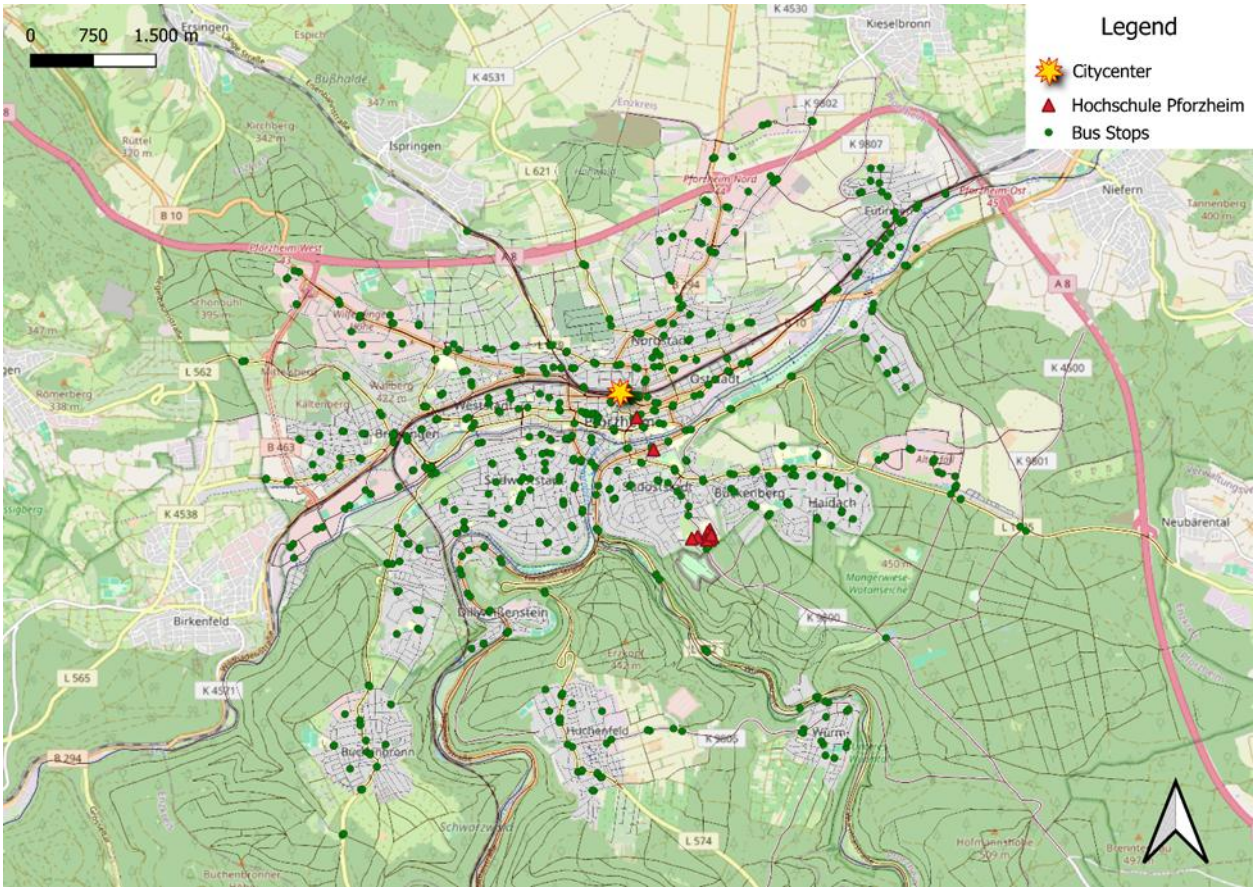
Pforzheim is a medium-sized city in the state of Baden-Württemberg, located at the northern edge of the Black Forest. It lies approximately 40 kilometers west of Stuttgart and functions as a regional hub for commerce, education, and mobility. The city covers a compact area of approximately 98 km^2, with a relatively balanced shape of around 12 km in both east–west and north–south directions (see Map 1 – Overview).


Legend
Pforzheim
~ 40 Km
0
5
10 km
Stuttgart
Ludwigsburg
Sindelfingen
Böblingen
Leonberg
Waiblingen
Esslingen a Neckar
Bietigheim-Bissingen
Vaihingen an der Enz
Mühlacker
Karlsbad
Bretten
Oberderdingen
Maulbronn
Königsbach-Stein
Remchingen
Birkenfeld
Niefern-Öschelbronn
Wiernsheim
Mönsheim
Tiefenbronn
Heimsheim
Rutesheim
Schwieberdingen
Möglingen
Markgröningen
Korntal-Münchingen
Ditzingen
Renningen
Weil der Stadt
Calw
Bad Wildbad
Schömberg
Neuhausen
Ostfildern
Filderstadt
Fellbach
Kernen im Remstal
Remseck am Neckar
Marbach am Neckar
Freiberg am Neckar
Steinheim an der Murr
Besigheim
Löchgau
Freudental
Erligheim
Bönnigheim
Gemmrigheim
Walheim
Mundelsheim
Großbottwar
Hessigheim
Ingersheim
Sachsenheim
Sersheim
Illingen
Eberdingen
Hemmingen
Weissach
Gerlingen
Leinfelden-Echterdingen
Denkendorf
Neuhausen auf den Fildern
Aidlingen
Gechingen
Althengstett
Ottenbach
Magstadt
Maichingen
Vaihingen
Bad Cannstatt
Untertürkheim
Obertürkheim
Steinenbronn
Neubulach
Bad Teinach-Zavelstein
Grafenau
Ispringen
Kieselbronn
Kämpfelbach
Eisingen
Keltern
Straubenhardt
Neuenbürg
Dobel
Hofen an der Enz
Bad Liebenzell
Simmozheim
Walzbachtal
Pfinztal
Gondelsheim
Knittlingen
Neulingen
Ölbronn-Dürn
Kieselbronn
Wurmberg
Ottenhausen
Ottenhausen
Ottenhausen

The *Map 7 - City Area* (see page 17) provides further details by overlaying the street network and highlighting the location of the city center. It visually supports the interpretation that accessibility in Pforzheim is strongly concentrated around the central areas, while peripheral districts, especially in the southern and southwestern parts appear more disconnected.

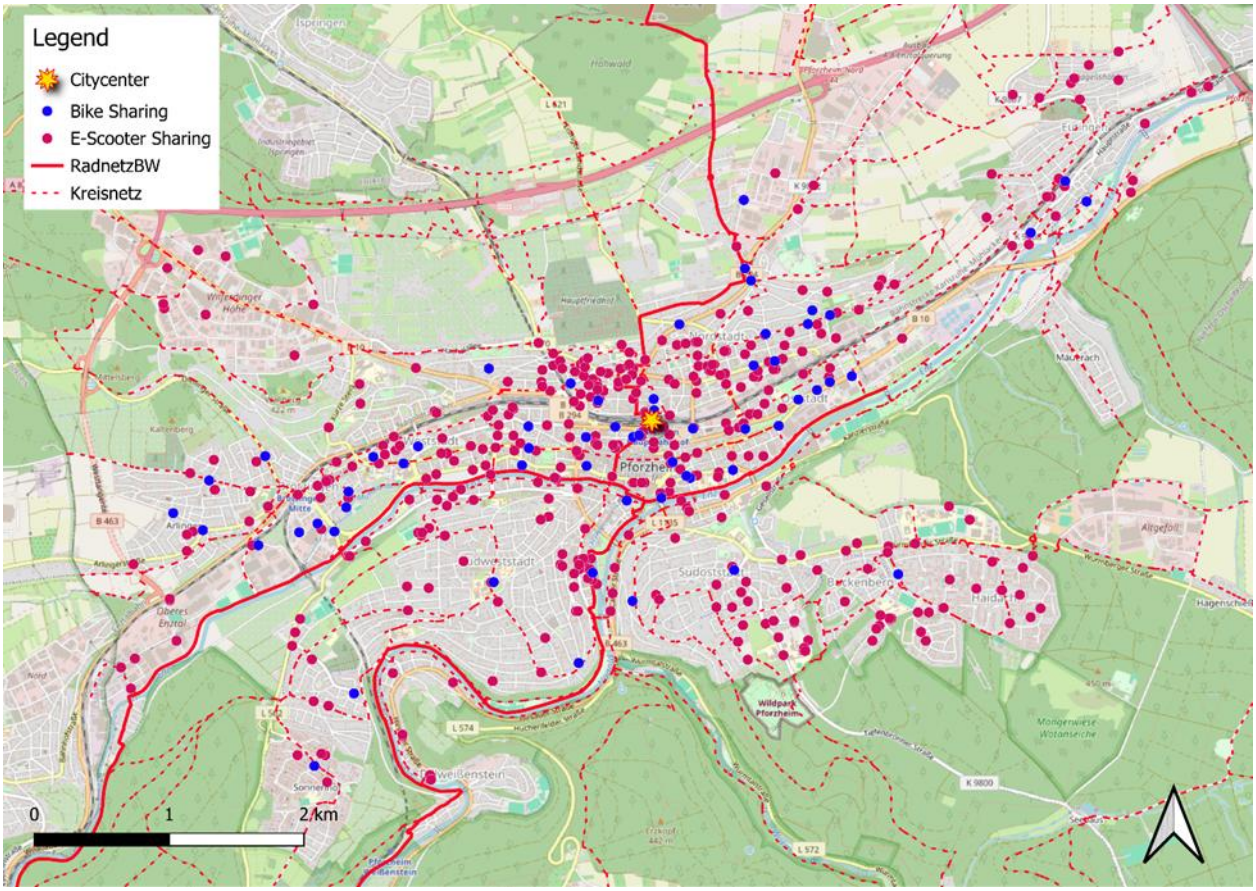
4 Mobility Infrastructure and Accessibility

The distribution of mobility infrastructure in Pforzheim reflects a relatively dense service provision in the inner city and more limited options in peripheral districts. The *Map 2 - Bus Stops* demonstrates a wide but uneven spread of public transport nodes. Bus stops are concentrated in the central corridor of the city and around the main train station, while outer districts, particularly in the southern and southwestern areas, show visibly lower stop density.



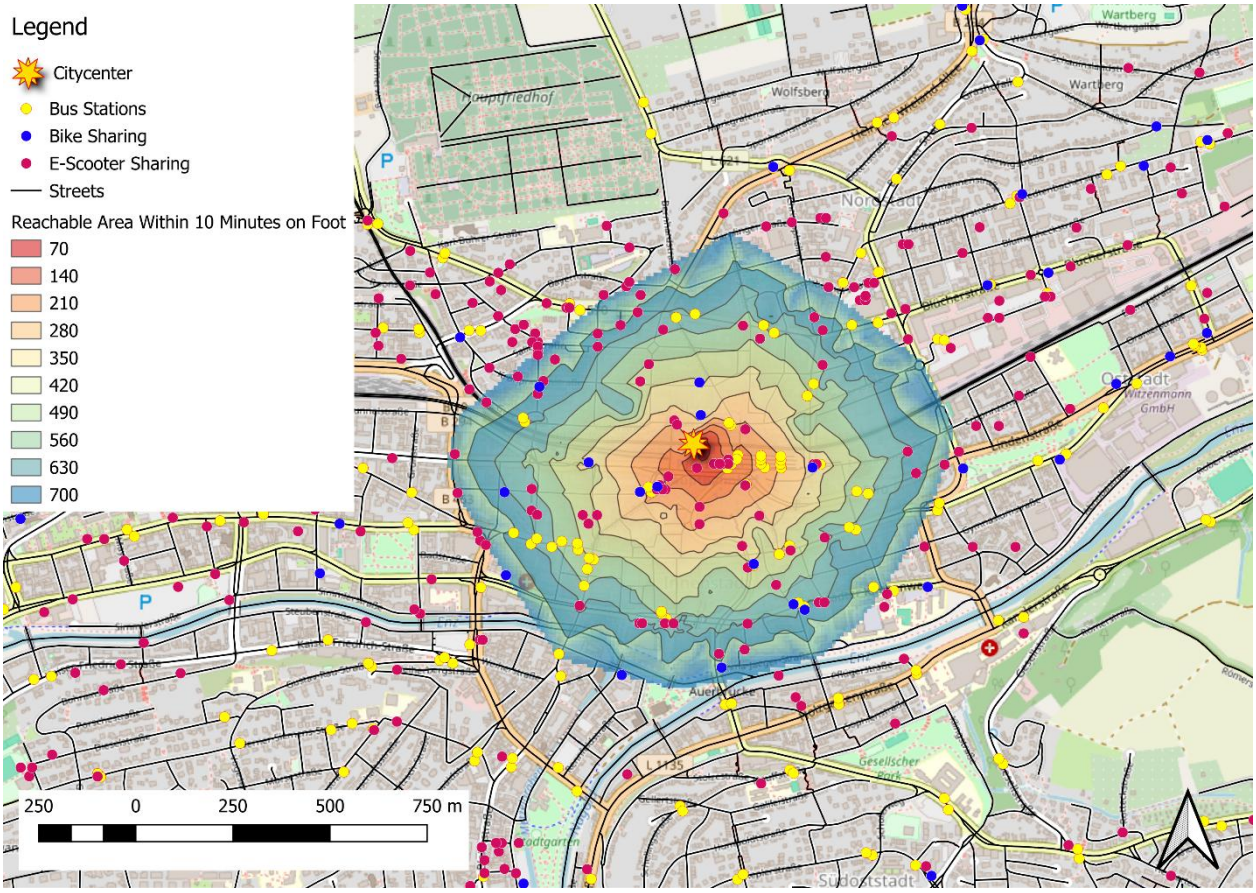
Map 2 - Bus Stops

In terms of shared micro-mobility, the *Map 3 - Bike & E-Scooter Sharing* illustrates the spatial distribution of both bike sharing and e-scooter sharing stations. As expected, the highest concentration of these services appears in the inner-city core, while large parts of the southern districts including areas around the Hochschule Pforzheim main campus remain underserved.



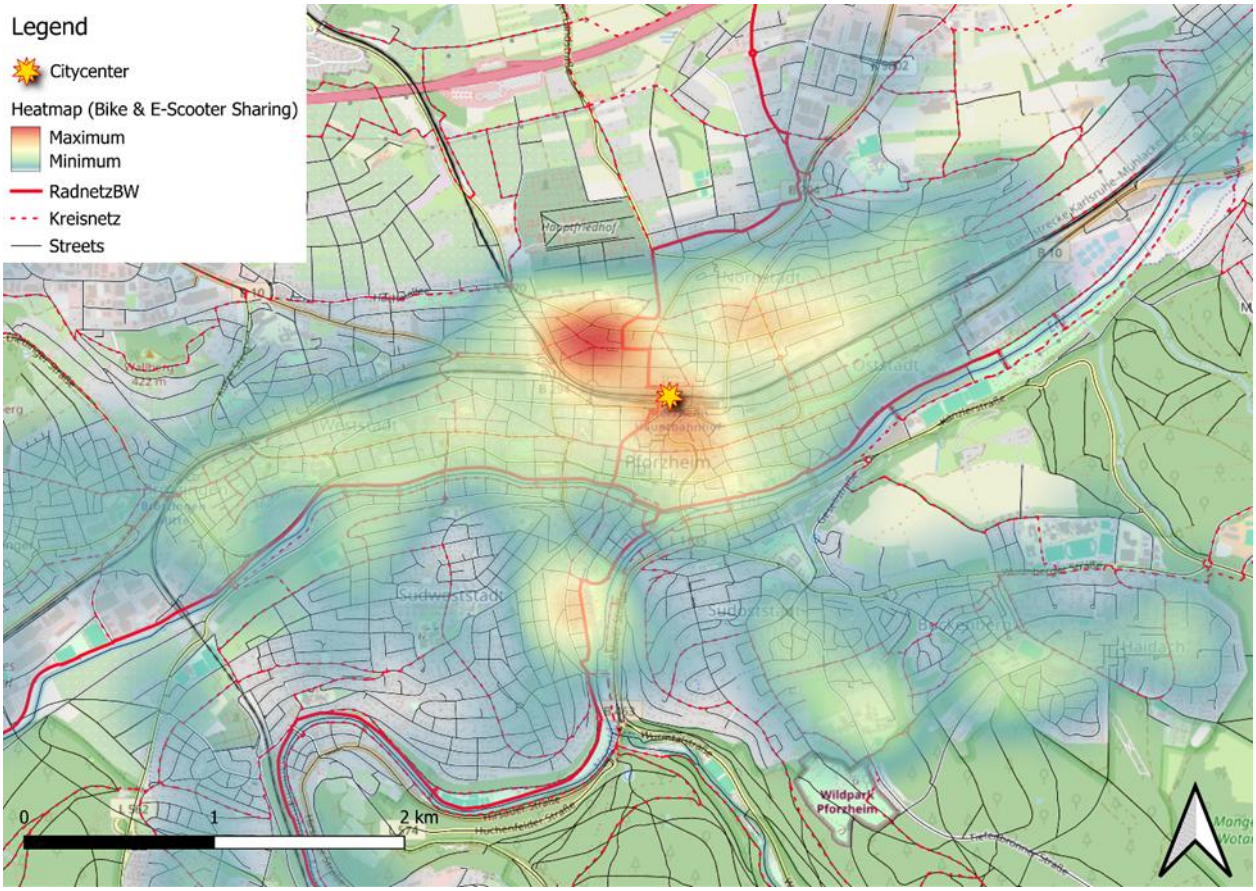
Map 3 - Bike & E-Scooter Sharing

The *Map 4 – 10 Minute Walk from City Center* visualizes the results of a 10-minute walking isochrone from the city center using a network-based accessibility analysis. Within this zone, a high number of mobility options are available, including multiple bus stops, bike sharing points, and e-scooter stations. This supports the assumption that multimodal accessibility is significantly better in the central area. Conversely, areas outside this zone lack the same level of accessibility, reinforcing existing mobility inequalities.



Map 4 – 10 Minute Walk from City Center

To complement this, the *Map 5 – Bike & E-Scooter Sharing Heatmap* clearly confirms the centralization of micro-mobility services. The strongest intensity of shared mobility usage and availability is concentrated in a few high-demand corridors near the center, tapering off almost entirely on the outskirts. Notably, the southern university area lies in a cold zone indicating either a lack of infrastructure or low demand due to disconnected placement.



Map 5 – Bike & E-Scooter Sharing Heatmap

While the city's core benefits from a strong clustering of multimodal options, the analysis reveals a significant drop in service density beyond the central urban ring. This spatial inequality has implications for both social equity and modal shift policies, especially in a city with relatively high private car usage.

According to *the infas, DLR, & BASt, 2019* data, Pforzheim shows the following modal split:

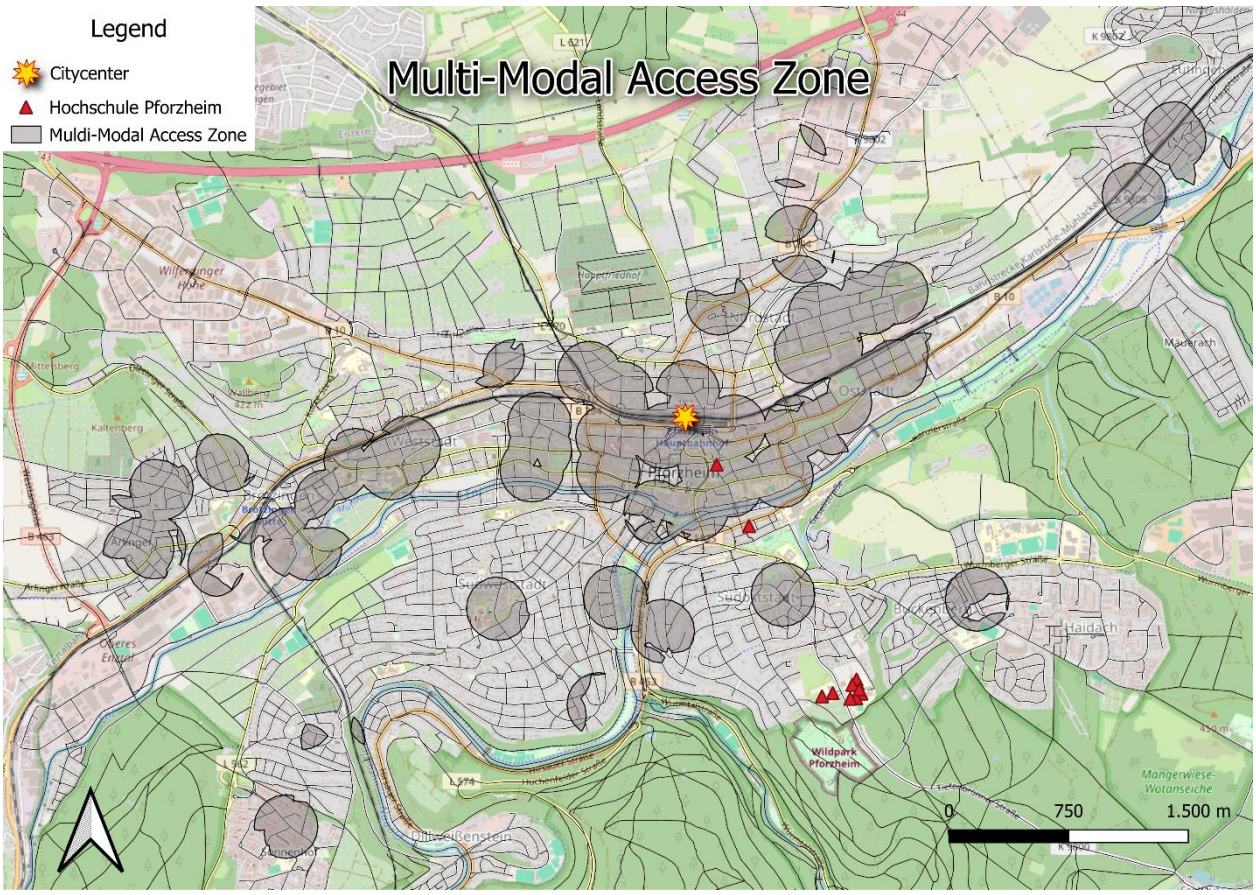
- Walking (zu Fuß): 25%
- Bicycle (Fahrrad): 6%
- Public transport (ÖV): 8%

- Motorized individual transport (MIV): 61%

o Of which drivers: 39%, passengers: 22%

This modal distribution positions Pforzheim as a car-dependent city compared to other medium-sized cities in the region. The relatively low share of cycling and public transport emphasizes the importance of improving accessibility and service availability beyond the central core *(infas, DLR, & BASt, 2019)*

In order to assess integrated mobility accessibility across the city, a multi-modal access zone analysis was conducted using a 200-meter buffer around three types of services: bus stops, bike sharing stations, and e-scooter sharing stations. The resulting *Map 6 – Multi-Modal Access Zone* highlights areas in which at least one of each three services is reachable within a 200-meter walking distance. (one Bus-Station, one E-Scooter Sharing and one Bike sharing)



Map 6 – Multi-Modal Access Zone

The output clearly shows that multi-modal accessibility is heavily clustered around the city center, forming a continuous zone of high service coverage. This area aligns with the location of the

central train station and major public and commercial institutions. As one moves toward the peripheral districts, especially toward the south and southwest, the presence of multi-modal zones becomes much more fragmented or even absent altogether.

A notable observation from this map is the lack of integrated mobility coverage in the southern part of the city, where Hochschule Pforzheim's main campus is located. Although the central campus area is relatively dense in buildings and student activity, it is not covered by any of the defined multi-modal zones. This suggests that within a 200-meter walking radius of these university buildings, there is no access to all three bus, bike sharing, and scooter sharing options. This is particularly striking given the campus's role as a regional educational hub, serving thousands of students and employees daily.

In contrast, the smaller university building located closer to the inner city identifiable in previous maps as one of the few red rectangles inside the city center mobility zone benefits from good accessibility and integration into the multi-modal network. This spatial contrast between campus locations underlines the unequal distribution of mobility options in Pforzheim.

The analysis supports the conclusion that Pforzheim's mobility infrastructure is strongly monocentric, favoring the city center, while outlying but socially important areas such as university campuses are left under-served. From a sustainable mobility planning perspective, this reveals a clear opportunity for improvement, especially considering the relatively low public transport share (8%) and bicycle usage (6%) reflected in the *infas, DLR, & BASt, 2019* modal split data.

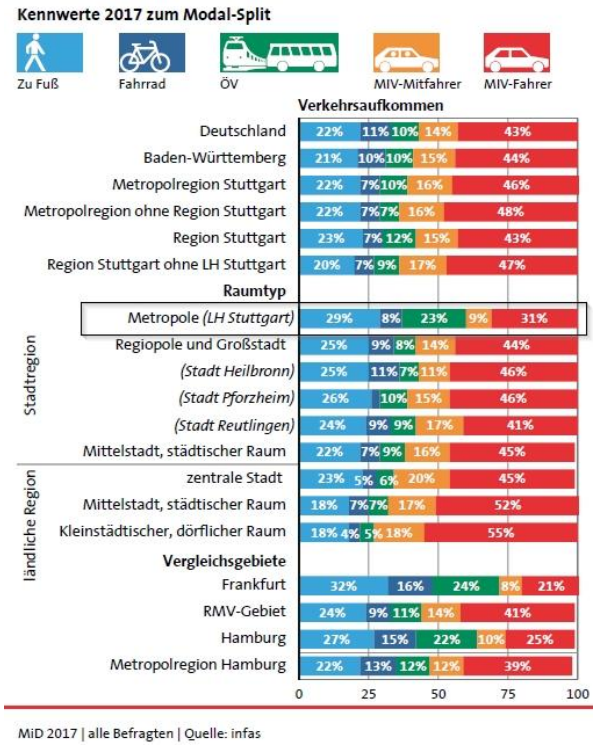


Figure 1 - Modal Split in Germany, Baden-Württemberg

5 Accessibility Challenges for Hochschule Pforzheim

The spatial evaluation of mobility services in relation to Hochschule Pforzheim reveals a clear disparity between the city center and the university's main campus cluster in the south. While the university is one of the most significant trip generators in Pforzheim, serving students, staff, and external visitors, its accessibility by shared and public mobility options remains suboptimal.

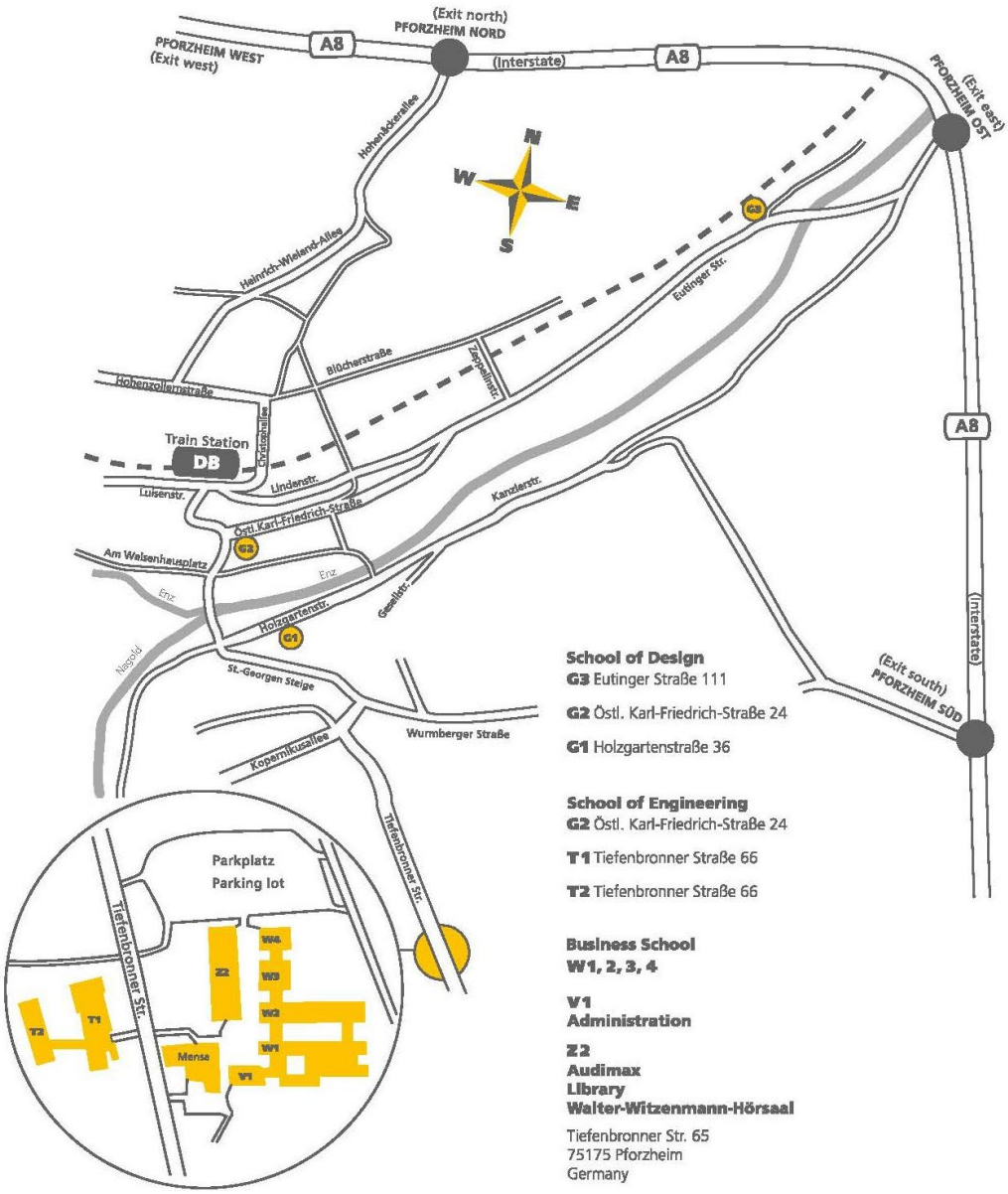


Figure 2 – Hochschule Pforzheim Campus Plan and Building Distribution

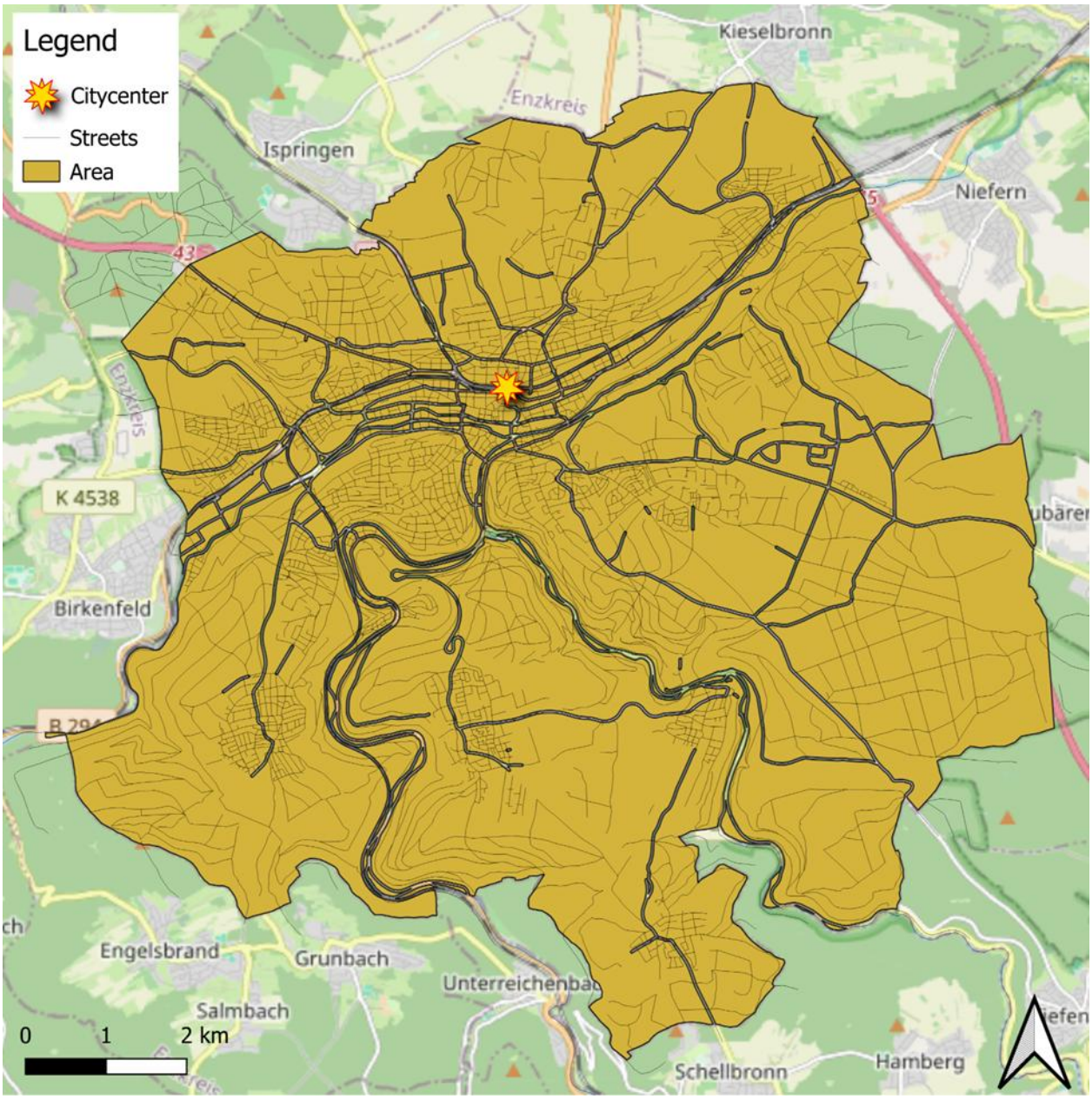
According to the official campus map provided by the university (Hochschule Pforzheim, 2024), the main academic buildings (T1, T2, W1–W4, G2, G3) are concentrated around Tiefenbronner

Straße in the city's southern districts. This area, while reachable by car and a few bus lines, lies outside the main multi-modal access zone, as shown in the Mobility Zone and Heatmap maps.
None of the southern university buildings fall within the 200-meter buffer zone where all three shared mobility options (bus, bike, scooter) are present. In fact, even single-mode access (e.g. scooter sharing) is very limited or completely absent. Furthermore, the Heatmap shows a visible cold spot around the campus, indicating either a lack of services or minimal user demand, possibly due to infrastructure gaps or the area's relatively isolated location.
While one smaller university building closer to the city center benefits from proximity to bike-sharing and bus services, this is an exception. The main student population spends their time in the south campus zone, which appears underserved from a sustainable mobility standpoint.
This mismatch between urban function (university = high activity node) and mobility supply reveals a structural weakness in the city's transportation planning. Given that students are a key target group for sustainable mobility strategies are often more flexible and more likely to use public or shared options. This gap limits potential mode shift.
Considering the *infas, DLR, & BASt, 2019* modal split, where private car usage (MIV) accounts for 61% of trips in Pforzheim, expanding the accessibility of multimodal options around the university could be an effective step toward encouraging a reduction in car dependency. Even small-scale additions, such as adding bike/scooter stations near the Menza and W-building cluster or increasing bus frequency could make a significant difference.

6 Mesh Size and Area Analysis

To better understand the internal spatial structure of Pforzheim, the city polygon was fragmented using the main road network, resulting in 102 distinct mesh segments. These mesh segments represent the smallest navigable zones or "city blocks" within the boundary.

The Area map shows the spatial distribution of these meshes, categorized by size. Each polygon represents a discrete fragment highlighting stark differences between compact inner-city blocks and expansive peripheral zones.



Map 7 – City Area

To quantify this fragmentation, the following statistics were calculated based on the area of each mesh:

Statistic	Value
Count	102
Sum	8,80E+07
Mean	862558
Median	73808,5
St dev (pop)	2,21E+06
St dev (sample)	2,22E+06
Minimum	2627
Maximum	1,47E+07
Range	1,47E+07
Variety	102
Q1	30420
Q3	333981
IQR	303561
Missing values	0

Table 1 – Area Statistics

These statistics confirm a high level of spatial inequality among mesh sizes:

- The large IQR (303,561 m^2) and the extremely wide range suggest that the urban structure contains both very dense areas and vast underdeveloped or open segments.
- The minimum area (2,627 m^2) likely corresponds to small blocks or edge-cuts, while the maximum (14.7 million m^2) probably represents parks, forests, or undeveloped peripheral land included in the polygon layer.

The combination of the visual Area map and the statistical table provides a powerful insight into the city's internal morphology, supporting further analysis of land use planning, service provision, and potential mobility inequality within zones.

7 Summary and Recommendations

This spatial analysis of Pforzheim's mobility infrastructure highlights both strengths and weaknesses in the city's current network of transportation options. Using open datasets and QGIS-based mapping, the study evaluated mesh structure, area fragmentation, shared mobility coverage, and university accessibility.

Key findings include:

- Centralized accessibility: Mobility services such as bike sharing, scooter sharing, and bus stops are clustered around the inner city, creating a compact multi-modal access zone. This confirms a mono-centric service pattern.
- Poor coverage in strategic zones: Despite being a major trip attractor, Hochschule Pforzheim's main campus is outside all high-access zones. The southern campus lacks sufficient scooter or bike-sharing infrastructure, with no coverage in the 200-meter multi-modal buffer zone.
- Fragmented mesh sizes: The road-based segmentation of the city resulted in 102 mesh zones, with a highly skewed area distribution. Large zones on the periphery contrast with small blocks in the city center, confirming a mixed-use and uneven urban pattern. This is visualized in the *Map 7 – City Area* and backed by detailed statistics (e.g., mean = 862,558 m^2, max = 14.7 million m^2).
- Mode choice patterns: Modal split data from the *infas, DLR, & BASt, 2019* shows Pforzheim's heavy reliance on private cars (61%), with minimal usage of public transport (8%) and bicycles (6%). This emphasizes the urgent need to invest in sustainable and inclusive mobility options especially in underserved areas.

Recommendations:

1. Add bike and scooter stations near Hochschule Pforzheim, especially around Tiefenbronner Straße and the Mensa/W1 area.
2. Introduce micro-mobility partnerships (e.g., with Voi or Tier) for the university, possibly under a student tariff.
3. Improve bus connections and frequency in the southern zone.
4. Use spatial analysis in future planning to ensure that areas with high population or functional density aren't excluded from the mobility network.

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